BioData Sage

Decode
2026

An Introduction to R and Data Wrangling with the Tidyverse

What is R?

- R is a programming language and environment for statistical computing and graphics
- It was developed in 1991
- R is both **open source** and **open development**

R can:

Manipulate data → Visualize → Perform statistics

Why R?

- Free and open source
- High level language
- Powerful and flexible
- Extensive add-on software (aka packages)
- Strong community!

What is RStudio?

- An Integrated Development Environment (IDE) for R
- Helps you:

Write code	View the output of your code	Find errors
Manage files	Interactive tools	View documentation

RStudio®

https://posit.co/downloads/

https://posit.co/products/open-source/rstudio

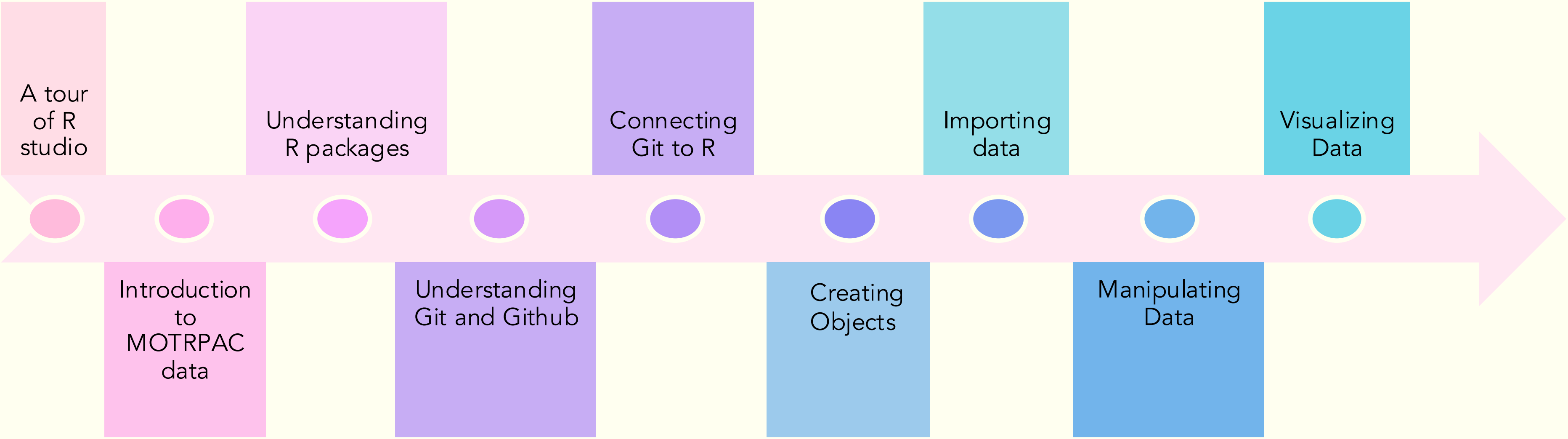
What is Posit Cloud?

- Platform that lets you use RStudio in a browser
- Removes the need to download software
- Lets you easily share projects for analysis and collaboration

https://posit.cloud/

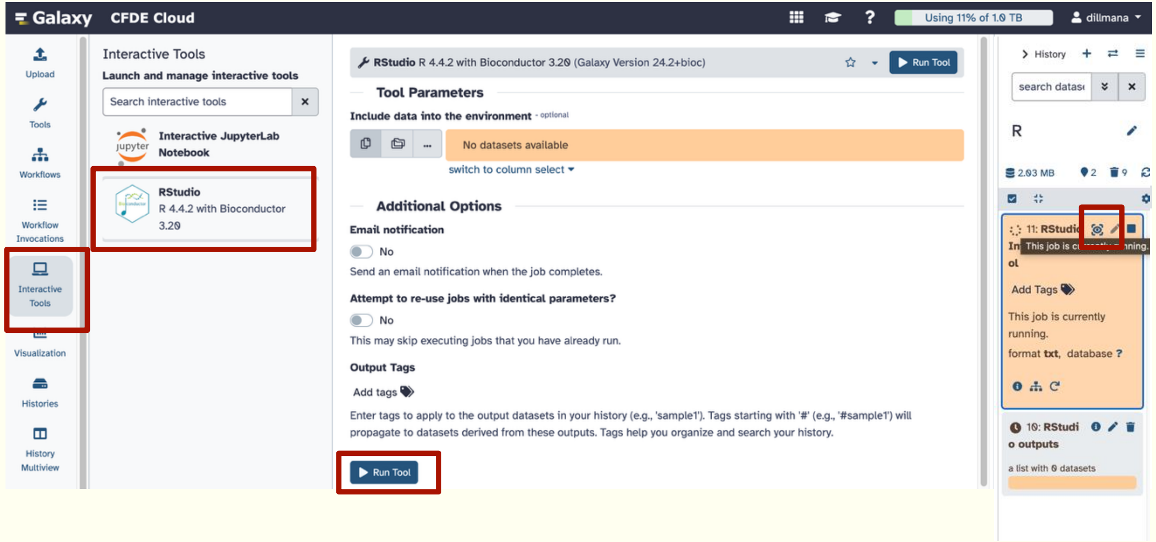
Outline

Data Manipulation and Visualization



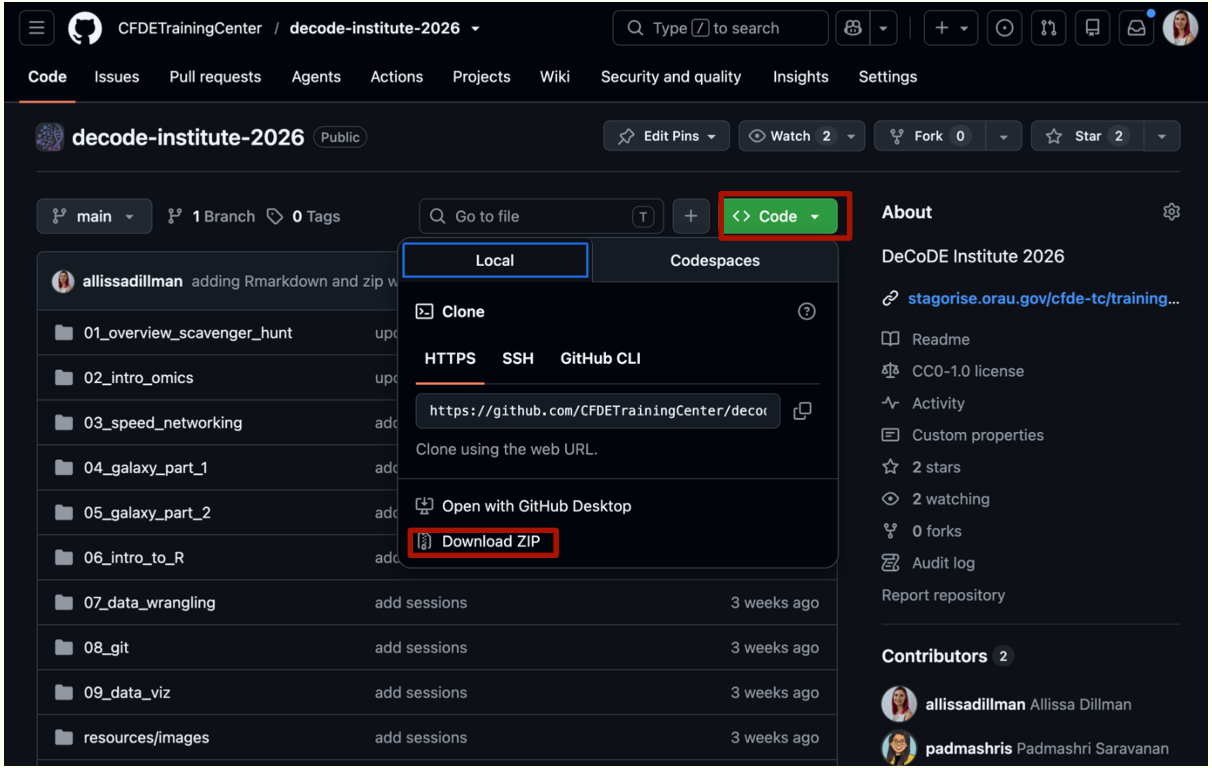
Let's get started!

- Log in to: https://cfdeworkspace.org/
- If you do not have an account this may take time a backup is https://usegalaxy.org/
- Click interactive Tools
- Click RStudio
- Click Run Tool
- Click View on the side panel to open RStudio



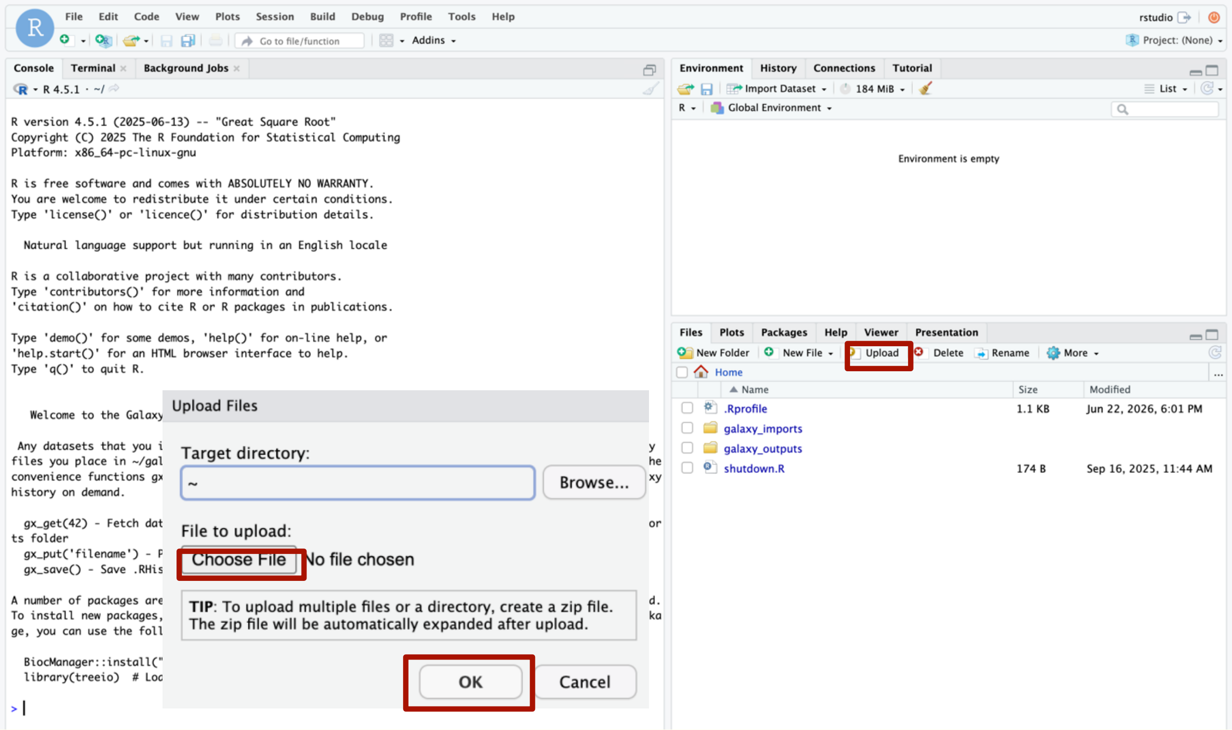
Let’s get started!

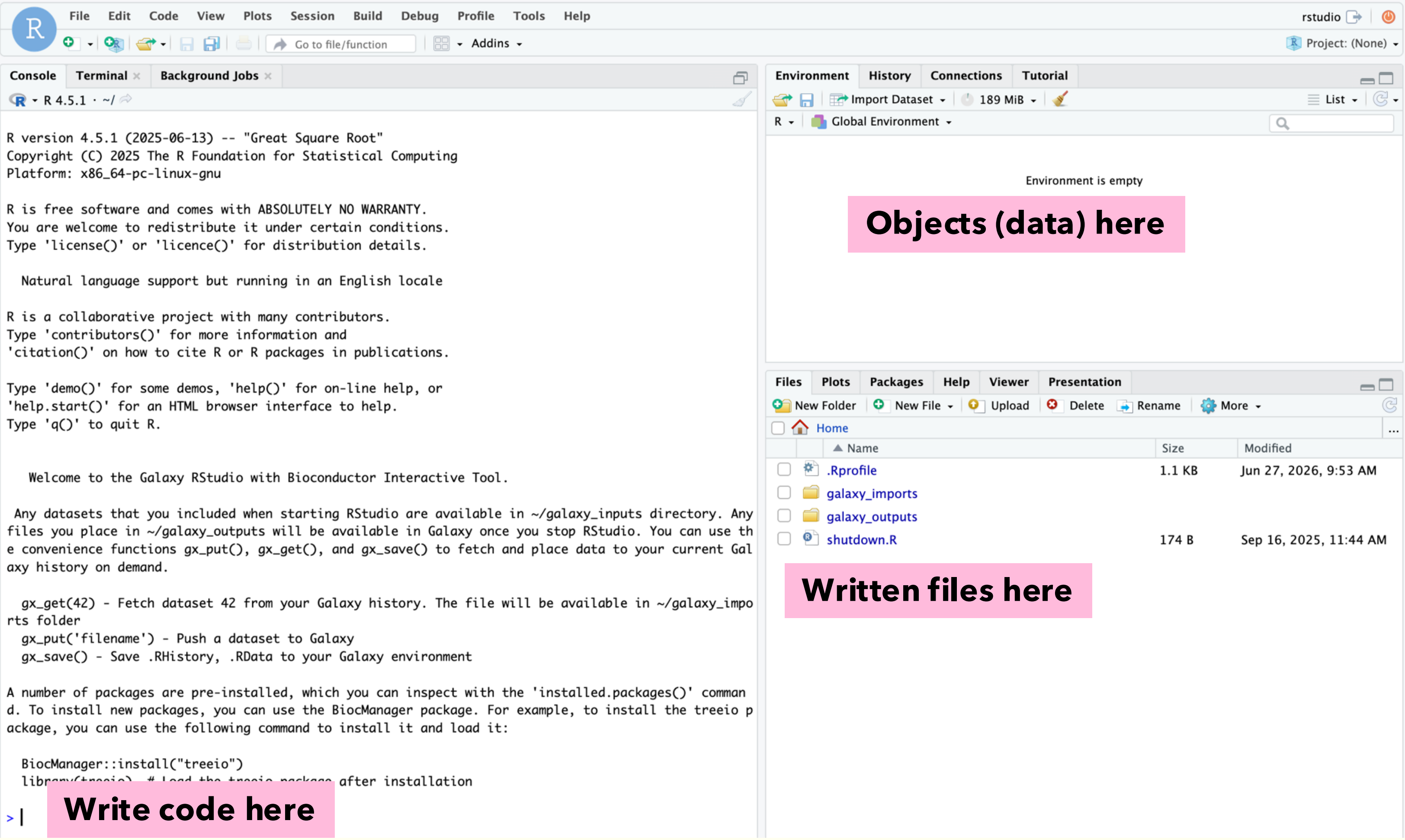
- https://github.com/CFDETrainingCenter/decode-institute-2026
- Click Code
- Click Download Zip



Let's get started!

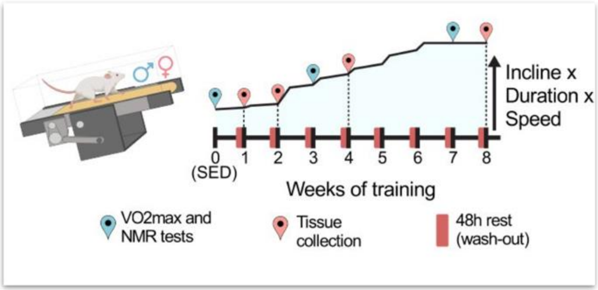
- Click on Upload
- Choose the Rmd file we downloaded from github
- Click Ok
- Repeat for zip file

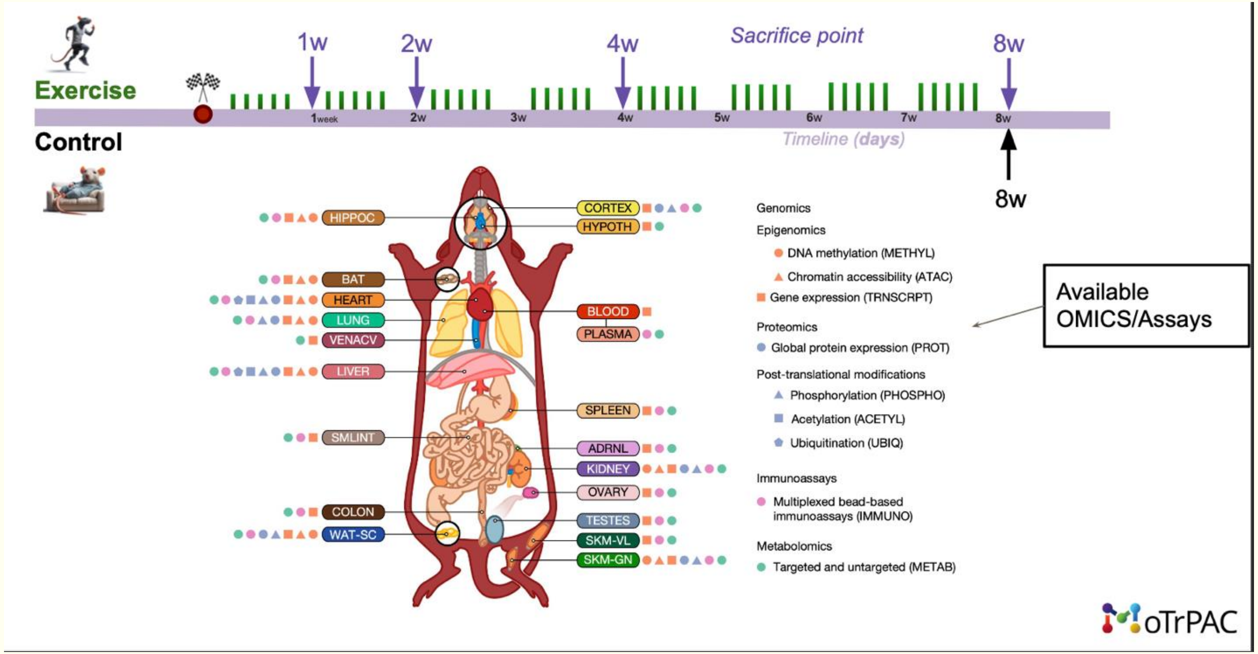




BioData Sage

What is MOTRPAC?

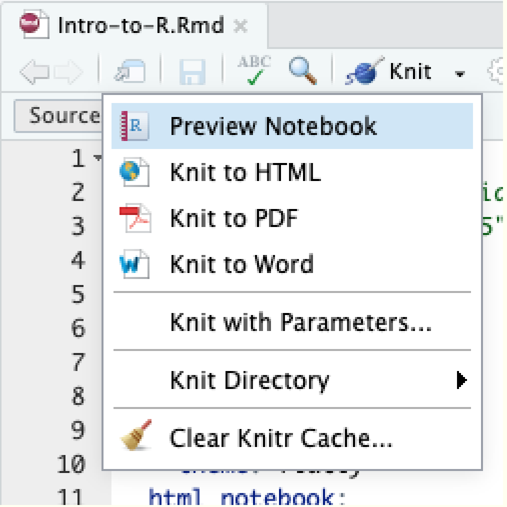




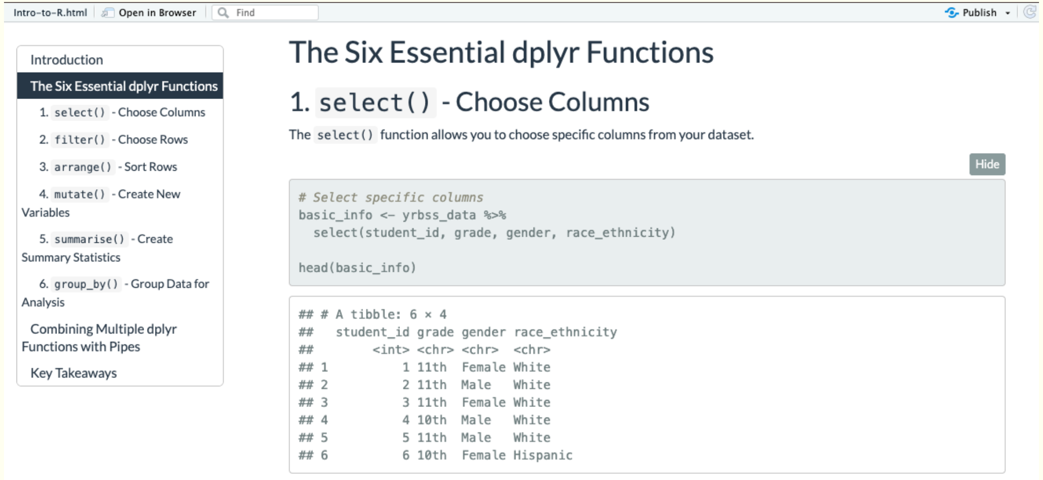
https://motrpac-data.org/

Interactive HTML files

Click on the dropdown and select *Knit to HTML*

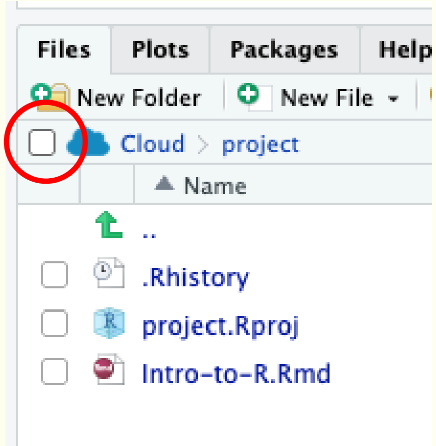


This will create a HTML file viewable in any browser without needing any coding platform

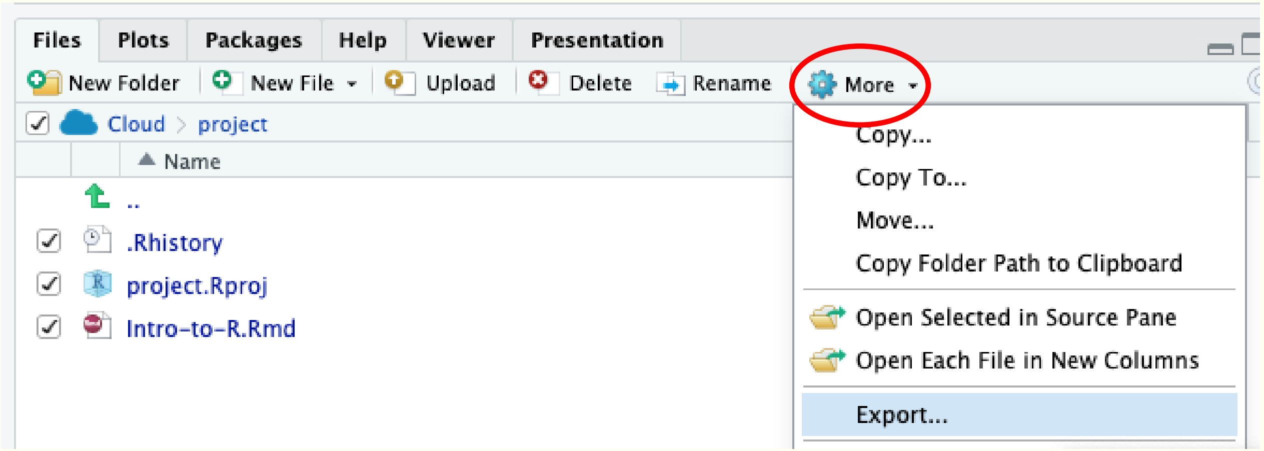


Exporting files out of the CWIC

Click on the checkbox on the left of the cloud icon on your *Files* pane



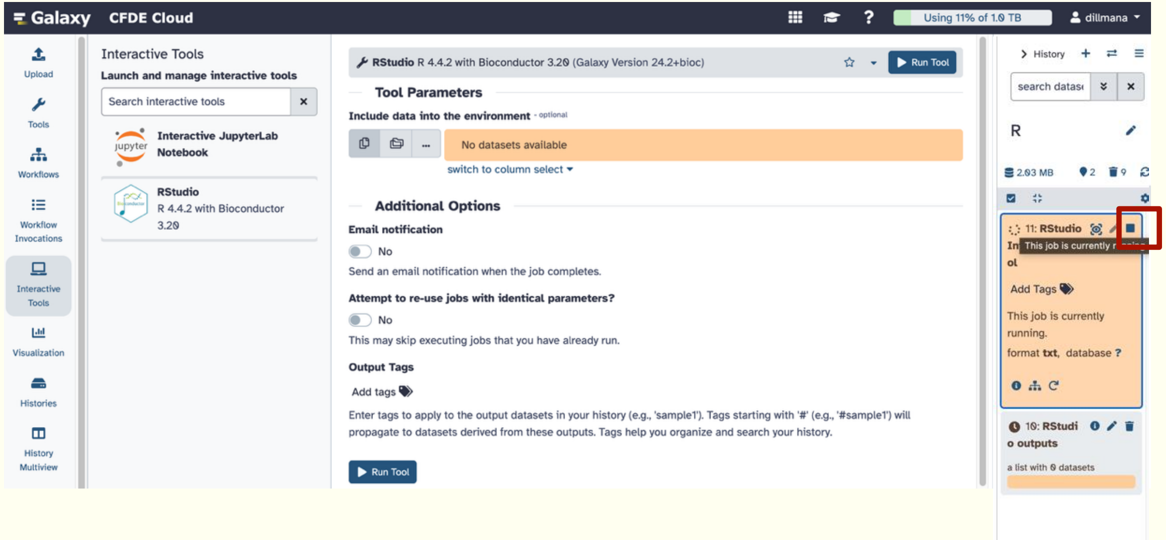
Select the *gear* icon / "More" in your *Files* pane, and select "Export" in the dropdown



You can unzip this file on your computer and click the .Rproj to open R with all the associated files automatically

Stopping Rstudio

- Go back to the Galaxy instance
- Click on the stop button on the Rstudio interactive tool



Q&A